Viscar Industrial Capacity Limited
Enhancing Performance & Productivity

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1. PLC Level 1 Course Outline (Beginner)

Duration: 1 Week (40 Hours)

Course Description

This course introduces participants to industrial automation using PLCs. It focuses on foundational PLC hardware, wiring, software tools, and basic programming. Emphasis is placed on hands-on practice using training simulators and industrial PLC platforms such as Siemens S7-1200/S7-300 and Schneider Electric PLC systems.

Course Content

Module 1: Introduction to PLC Systems
(Theory-5hrs, practical-3hrs)

Industrial automation overview
Relay logic vs PLC control
PLC architecture and components
PLC scan cycle
Basic electrical symbols

Module 2: PLC Programming Fundamentals
(Theory-5hrs, practical-3hrs)

IEC 61131-3 overview
Ladder logic fundamentals
Contacts, coils, and memory bits
Creating and downloading PLC programs

Module 3: Timers and Counters
(Theory-5hrs, practical-3hrs)

TON, TOF, pulse timers
Up/Down counters
Industrial applications

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Module 4: PLC Wiring and Troubleshooting

(Theory-5hrs, practical-3hrs)

Sinking vs sourcing inputs
Sensor wiring (PNP/NPN)
Protection devices and wiring interpretation
PLC fault identification and forcing I/O

Module 5: Practical Mini-Project

(Theory-3hrs, practical-5hrs)

Motor start/stop system
Tank level simulation
Traffic light logic

Specific Learning Outcomes

By the end of this course, participants will be able to:

- Identify PLC hardware components and functions
- Wire basic PLC input and output devices safely
- Develop and download simple PLC programs
- Apply timers and counters in control applications
- Interpret basic wiring diagrams
- Diagnose and troubleshoot simple PLC faults
- Implement small automation projects independently

Target Audience

Electrical, Mechatronics, and Mechanical Engineers & Technicians